

## Career Summary

Final-year Bachelor of Science in Telecommunication and Information Engineering student at Dedan Kimathi University of Technology with practical exposure to fiber-optic engineering, telecommunications systems, software development, data workflows, and artificial intelligence. Hands-on experience building simulation-based IoT systems, machine-learning solutions, web applications, and engineering models using Python, MATLAB/Simulink, Proteus, Firebase, PyTorch, and Flask. Also experienced in data annotation, data labeling, AI training data preparation, dataset curation, and model evaluation. Combines engineering problem-solving with growing data and AI capabilities to develop practical technology solutions.

## Expertise

Telecommunications & Fiber Optics | Data Engineering Fundamentals | Data Annotation & Labeling | AI Training Data Preparation | Machine Learning | IoT Systems | Signal Processing | System Simulation | Software Development | Technical Troubleshooting | Research & Technical Documentation

## Technical Skills

Python | Java | JavaScript | HTML/CSS | MATLAB & Simulink | PyTorch | TensorFlow concepts | Flask | Firebase Realtime Database | NoSQL | Chroma/Vector Stores | Git & VS Code | Proteus | Data Cleaning & Preparation | Dataset Curation | Data Labeling & Annotation | AI/ML Model Evaluation | Digital Communications | Fiber Optics | Electrical Installations | GIS

## Professional Experience

**Fiber Optic Engineer Attachee | Agape Enterprise Solutions | Nyeri, Kenya** Jan 2025 – Apr 2025

- Supported practical fiber-optic engineering and ISP operations, gaining exposure to real-world telecommunications infrastructure and technical support.
- Applied engineering principles to fiber-optic installation, troubleshooting, connectivity support, and technical documentation.
- Worked with technical staff to diagnose problems and strengthen practical communication, teamwork, and field problem-solving skills.

**Engineering Trainee | Academic & Laboratory Training | Dedan Kimathi University of Technology** 2022 – Present

- Developed and tested telecommunications, control, signal-processing, and digital-communication systems using MATLAB, Simulink, and circuit simulation tools.
- Performed engineering analysis, system modeling, simulation, validation, and technical documentation across multiple undergraduate engineering units.
- Applied programming and software-development skills to engineering and data-oriented projects, including Python-based backend and data workflows.

## Education

**Bachelor of Science | Telecommunication and Information Engineering | Dedan Kimathi University of Technology** 2022 – Present

- Final-year undergraduate with practical and academic work spanning telecommunications, digital communications, signal processing, networking, control systems, GIS, antennas, satellite communications, and software/data engineering.

**KCSE | Baringo High School** 2017 – 2022 | Grade: B+

## Interests

Artificial Intelligence & Data | Telecommunications & Networking | Software Development | IoT | Emerging Technologies | Engineering Innovation | Technical Research

## Skills

**Data & AI:** Data cleaning and preparation | Data annotation | Data labeling | AI training data preparation | Dataset curation | Machine learning workflows | Model evaluation | NLP and intent classification

**Data Engineering:** Data preprocessing | Structured data handling | NoSQL concepts | Firebase Realtime Database | Data pipelines and serial data workflows | Vector-store concepts

**Programming:** Python | Java | JavaScript | HTML/CSS | MATLAB | Flask | Git | VS Code

**Machine Learning:** PyTorch | EfficientNet | ResNet | TensorFlow concepts | Image classification | Training and validation workflows

**Telecommunications:** Fiber optics | Digital communications | Signal processing | Transmission lines | Antennas | Satellite communications | Control systems | Telecommunications system simulation

**Engineering & Simulation:** MATLAB/Simulink | Proteus | Circuit simulation | System modeling | Electrical installations | GIS

**Professional:** Analytical problem-solving | Technical documentation | Research | Troubleshooting | Team collaboration | Communication | Adaptability